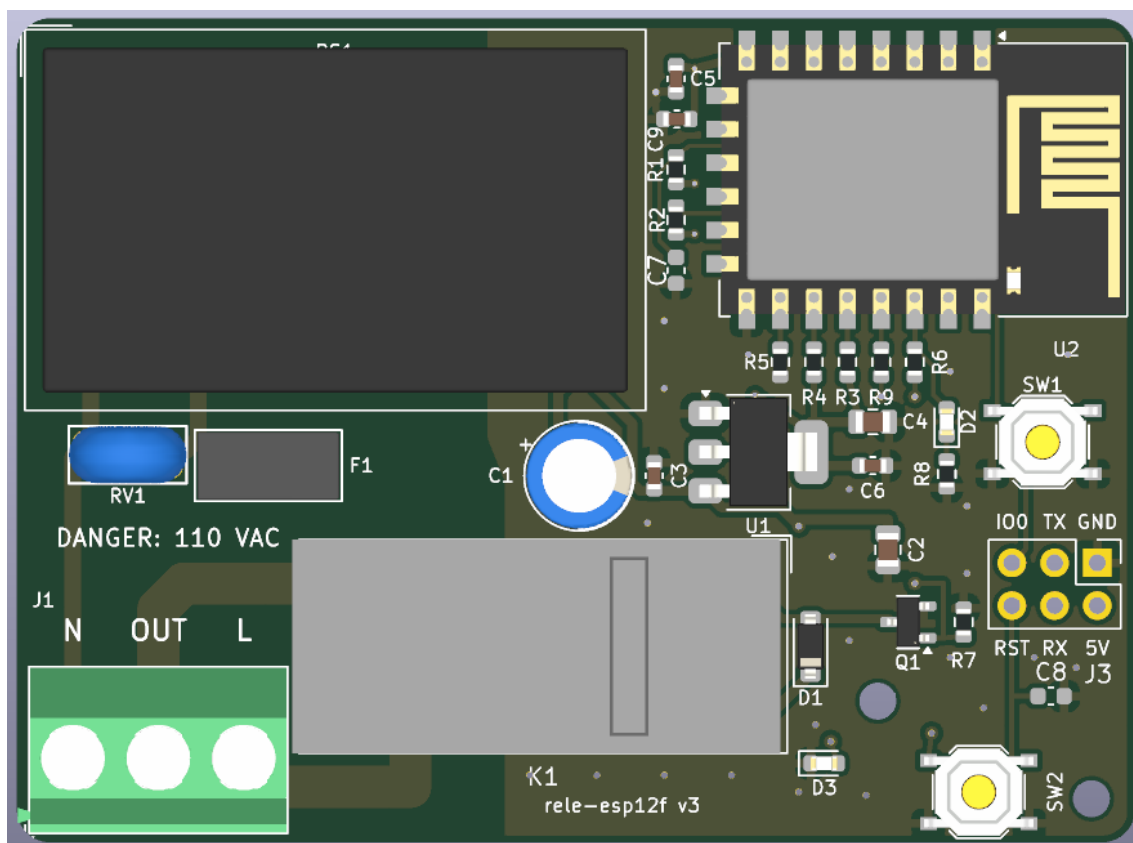


rele-esp12f

Single-channel Wi-Fi relay module for 110–120 VAC

ESP-12F (ESP8266EX) · Omron G5RL-1A-E-HR 16 A · HLK-PM01 isolated supply

Hardware v3 · Document rev. B · 2026-10-04



- **Mains switching:** 1 normally-open contact (SPST-NO) on the line conductor, 10 A resistive (board limit); 16 A relay, high-inrush model (100 A).
- **Integrated supply:** direct 110–120 VAC input, HLK-PM01 isolated 5 V / 3 W supply, 3.3 V LDO.
- **2.4 GHz Wi-Fi** 802.11 b/g/n with ESP8266EX; compatible with open firmware (Tasmota, ESPHome, Arduino).
- **Protection:** T500 mA time-lag fuse for the supply, 07D221K varistor, flyback diode on the coil.
- **Isolation:** ≥ 4 mm in copper between mains and low voltage; 8 mm coil-to-contact in the relay (reinforced insulation).
- **Full assembly at JLCPCB:** 32 parts, single side; 66.5 × 49 mm, 2 layers.

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Part I

Technical datasheet

Electrical, functional and design specifications of the rele-esp12f module, hardware revision v3.

How to read this document. **Part I** is the datasheet: limits, characteristics, functional description and design rationale of the board. **Part II** is the user guide: safety, firmware flashing, installation, configuration, operation, maintenance and troubleshooting. Every numeric value in the specification tables carries a source tag, because not all of them have the same degree of certainty:

Tag	Meaning
[D]	Manufacturer datasheet included in docs/datasheets/ (Omron G5RL, AMS1117, ESP8266EX).
[F]	Manufacturer data whose datasheet is not in the project (e.g. HLK-PM01, KANGNEX terminal block, varistor). Verify before relying on it.
[C]	Calculated from the design (schematic, layout, IPC-2221). The calculation is in Appendix A.
[M]	Measured on the KiCad layout (distances, dimensions). Not a measurement on a physical board.
[E]	Design estimate or recommendation. Not measured on a real board.

1. General description

The **rele-esp12f** is a single-channel, Wi-Fi-controlled mains switching module. It connects directly to a 110–120 VAC line with neutral, is powered from it through an isolated switch-mode supply and drives a load of up to 10 A through an Omron power relay whose normally-open contact is in series with the line conductor. Control is handled by an ESP-12F module (ESP8266EX SoC), which can run open firmware such as Tasmota or ESPHome, or custom code.

The board is physically split into two zones: the **mains zone** (terminal block, fuse, varistor, supply input and relay contacts) and the extra-low-voltage **SELV zone** (5 V and 3.3 V, microcontroller, driver, LEDs, push buttons and programming header). The two zones only interact through two parts that provide their own isolation: the HLK-PM01 supply and relay K1.

1.1 Key features

- Mains input 110–120 VAC, 50/60 Hz, with neutral. Estimated self-consumption ≈ 0.6 W idle and ≈ 1.2 W with the relay closed [E].
- Switched-line SPST-NO output; 10 A rms resistive as the board limit; Omron G5RL-1A-E-HR relay rated 16 A / 250 VAC, inrush up to 100 A [D].
- Hi-Link HLK-PM01 isolated supply (5 V, 600 mA) protected by a T500 mA fuse and a 140 VAC varistor [F].
- AMS1117-3.3 regulator and 10 μ F + 100 nF decoupling next to the ESP-12F, following the Espressif guidelines.
- ESP8266EX: Xtensa L106 CPU at 80/160 MHz, Wi-Fi 802.11 b/g/n, +20 dBm in 802.11b [D].
- Safe boot state: the relay stays open during reset and boot (pull-down resistor on the transistor base).
- Indicators: red relay-status LED (hardwired) and blue Wi-Fi LED (firmware-controlled).
- Local push button SW1 (GPIO0, also used to enter flashing mode) and reset push button SW2.
- Programming header J3 (2×3, 2.54 mm) powered at 5 V from the USB-serial adapter; later updates over the air (OTA).
- 2-layer FR-4 PCB, 1.6 mm, 35 μ m copper, 66.5 × 49 mm. All components on the top side.

1.2 Applications

- Remote switching of lighting, fans, resistive heaters and small appliances.
- Home-automation integration (Home Assistant, MQTT) with Tasmota or ESPHome.
- Timed and scheduled control of mains loads.

1.3 Block diagram

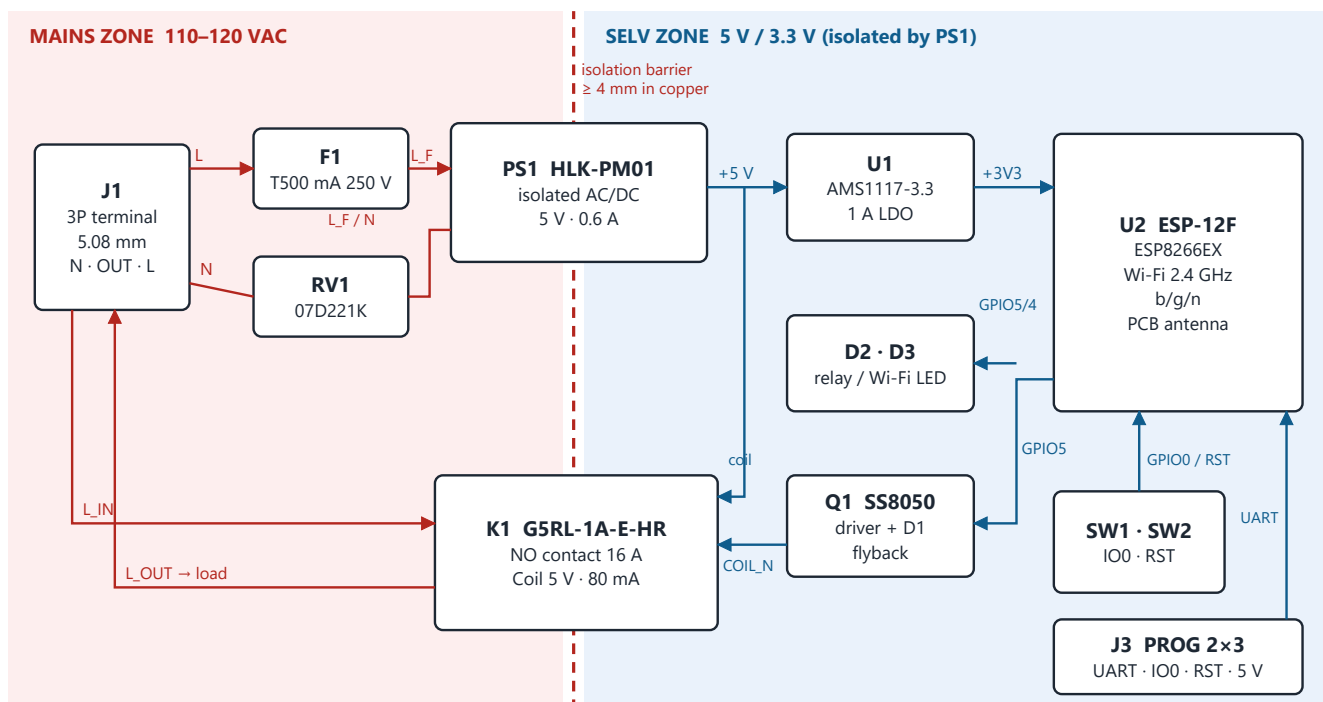


Figure 1. Block diagram. The red dashed line is the isolation barrier: only PS1 (supply transformer) and K1 (8 mm coil-to-contact separation) cross it.

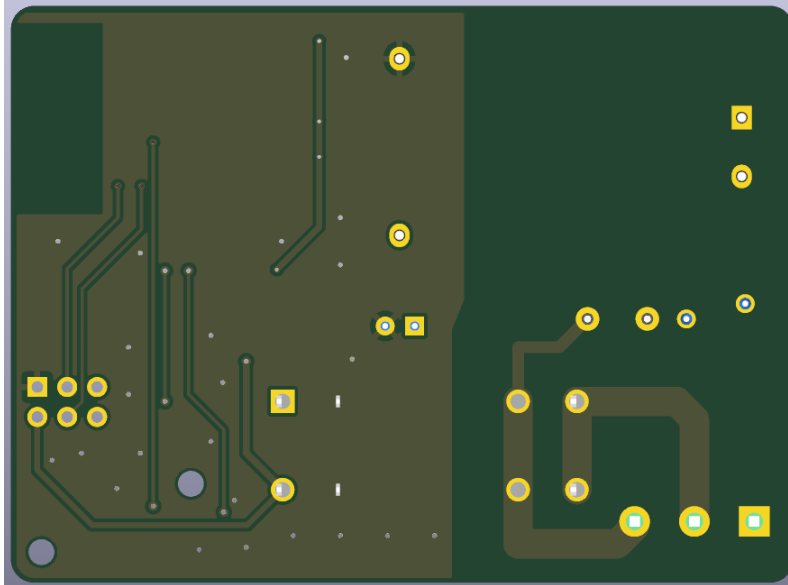


Figure 2. 3D render of the bottom side: copper, solder mask and THT leads only; there are no components (single-sided assembly).

1.4 Product status

Development prototype, not certified

- This board has not undergone electrical safety (IEC 62368-1, UL, CE) or electromagnetic compatibility testing. Isolation distances follow a design criterion (4 mm), not a formal assessment.
- Espressif lists the ESP8266EX as **NRND** (not recommended for new designs) in its 2025.11 datasheet. The module is still available, but its replacement should be planned for future revisions.
- Values tagged [E] have not been measured on a physical board. See Appendix C for known open points.

2. Specifications

2.1 Absolute maximum ratings

Exceeding any of these values may permanently damage the board or create a fire or shock hazard. They are not operating conditions.

Parameter	Condition	Limit	Source / reason
Mains voltage L–N	continuous	140 VAC rms	[F] Maximum continuous voltage of the 07D221K varistor. Above it, the varistor conducts and is destroyed.
Load current (J1 OUT)	continuous rms	10 A	[C] 2×2.5 mm traces, $35 \mu\text{m}$: $+12^\circ\text{C}$ at 10 A (IPC-2221).
Inrush peak current	K1 contact	100 A peak	[D] VDE test of the G5RL-1A-E-HR: 240 VAC, 100 A (0-P), 10 A steady state.
Contact voltage	K1	250 VAC	[D] Omron G5RL.
Voltage on J3 pin 2 (+5 V)	mains disconnected	5.5 V	[E] K1 coil ≤ 6.5 V (130 %) [D]; C1–C3 rated 10 V; margin over USB (5.25 V).
Voltage on TX, RX, IO0, RST (J3)	—	$-0.3 \dots 3.6$ V	[D] ESP8266EX, V_{IH} max. 3.6 V. Not 5 V tolerant.
Current per GPIO	per pin	12 mA	[D] ESP8266EX, I_{MAX} .
U1 junction temperature	—	125°C	[D] AMS1117.
Electrostatic discharge	HBM, ESP pins	2 kV	[D] ESP8266EX. Handle J3 with ESD precautions.

2.2 Recommended operating conditions

Parameter	Min.	Typ.	Max.	Unit	Notes
Mains voltage	100	115	127	VAC	110/120 V mains $\pm 10\%$. For 230 V, RV1 must be replaced (e.g. 07D471K).
Mains frequency	47	50/60	63	Hz	[F] HLK-PM01.
Resistive load	—	—	10	A	See §2.4 for other load types.
Ambient temperature (inside the enclosure)	0	25	40	$^\circ\text{C}$	[E] Conservative recommendation. The relay is rated $-40\dots 85^\circ\text{C}$ [D]; the real limit is set by PS1 and C1 [F].
Relative humidity	5	—	85	%	[D] Omron G5RL; non-condensing.
Voltage on J3 (flashing only)	4.75	5.0	5.25	V	VBUS from the USB-serial adapter.

2.3 Electrical characteristics

Unless otherwise noted, at 25°C and 115 VAC.

Parameter	Symbol / condition	Min.	Typ.	Max.	Unit	Source
Power supply						
5 V rail voltage	V_{5V} , PS1 output	—	5.0	—	V	[F]
Available current on 5 V	I_{PS1}	—	—	600	mA	[F]
3.3 V rail voltage	V_{3V3} , 0–0.8 A, full T_j	3.201	3.300	3.399	V	[D]
U1 dropout	$I_{OUT} = 0.8$ A	—	1.1	1.3	V	[D]
Available regulation headroom	$V_{5V} - V_{3V3}$	—	1.7	—	V	[C]
ESP8266EX current, average	connected to Wi-Fi	—	80	—	mA	[D]
ESP8266EX current, transmitting	802.11b, 11 Mbps, +17 dBm	—	170	—	mA	[D]
K1 coil current	5 V	—	80	—	mA	[D] $\pm 10\%$
Total load on the 5 V rail	relay closed + continuous TX	—	≈ 266	—	mA	[C] 44 % of PS1
U1 dissipation	average / TX peak	—	0.14 / 0.29	—	W	[C]
Power drawn from mains	relay open / closed	—	$\approx 0.6 / 1.2$	—	W	[E] PS1 $\eta \approx 70\%$
Output (relay K1)						
Contact configuration	—	SPST-NO (1 NO), switches the line conductor				
Load current, board limit	resistive, rms	—	—	10	A	[C]
Contact rated current	250 VAC resistive	—	—	16	A	[D]

Parameter	Symbol / condition	Min.	Typ.	Max.	Unit	Source
Contact resistance	1 A, 5 VDC, initial	—	—	100	mΩ	[D]
Operate time	—	—	—	15	ms	[D]
Release time	without diode; D1 extends it	—	—	5	ms	[D]
Coil must-operate / must-release voltage	% of 5 V	≥ 0.5 V	—	≤ 3.5 V	V	[D] 70 % / 10 %
Voltage across the coil	5 V – $V_{CE(sat)}$	—	≈ 4.8	—	V	[C]
Electrical endurance	16 A, 250 VAC resistive, 1,800 ops/h	50,000	—	—	ops	[D]
Mechanical endurance	18,000 ops/h	10 ⁷	—	—	ops	[D]
Isolation						
Mains ↔ SELV copper clearance	minimum, both sides	4.35	—	—	mm	[M] L_IN ↔ GND plane
Mains copper ↔ board edge	minimum	2.0	—	—	mm	[M] J1 N pad
K1 coil ↔ contact	clearance / creepage	8 / 8	—	—	mm	[D] reinforced
Coil ↔ contact dielectric strength	1 min, 50/60 Hz	6,000	—	—	VAC	[D]
Coil ↔ contact impulse	1.2 × 50 μs	10	—	—	kV	[D]
PS1 input–output isolation	—	see Hi-Link datasheet				[F]

Parameter (microcontroller and radio)	Condition	Value	Source
SoC	—	ESP8266EX, Xtensa L106 32-bit, 80/160 MHz	[D]
Operating voltage	—	2.5 ... 3.6 V (3.3 V typ.)	[D]
Flash memory	ESP-12F module	4 MB typ. (depends on module batch/vendor)	[F]
Band	—	2,412 ... 2,484 MHz, 802.11 b/g/n (HT20)	[D]
Transmit power	802.11b / g / n	+20 / +17 / +14 dBm	[D]
Sensitivity	11 Mbps CCK / 54 Mbps / MCS7	–91 / –75 / –72 dBm	[D]
Logic levels	V_{IL} / V_{IH} / V_{OH}	≤ 0.25 V_{IO} / ≥ 0.75 V_{IO} / ≥ 0.8 V_{IO}	[D]
Enable delay	R1·C7 = R2·C8	τ = 1 ms (EN reaches 0.75· V_{IO} in ≈ 1.4 ms)	[C]

2.4 Switching capacity by load type

The G5RL-1A-E-HR relay is designed for high-inrush loads, but the board limits the steady-state current to 10 A. For non-resistive loads, contact wear depends on the make inrush and on the break arc, so derating is recommended:

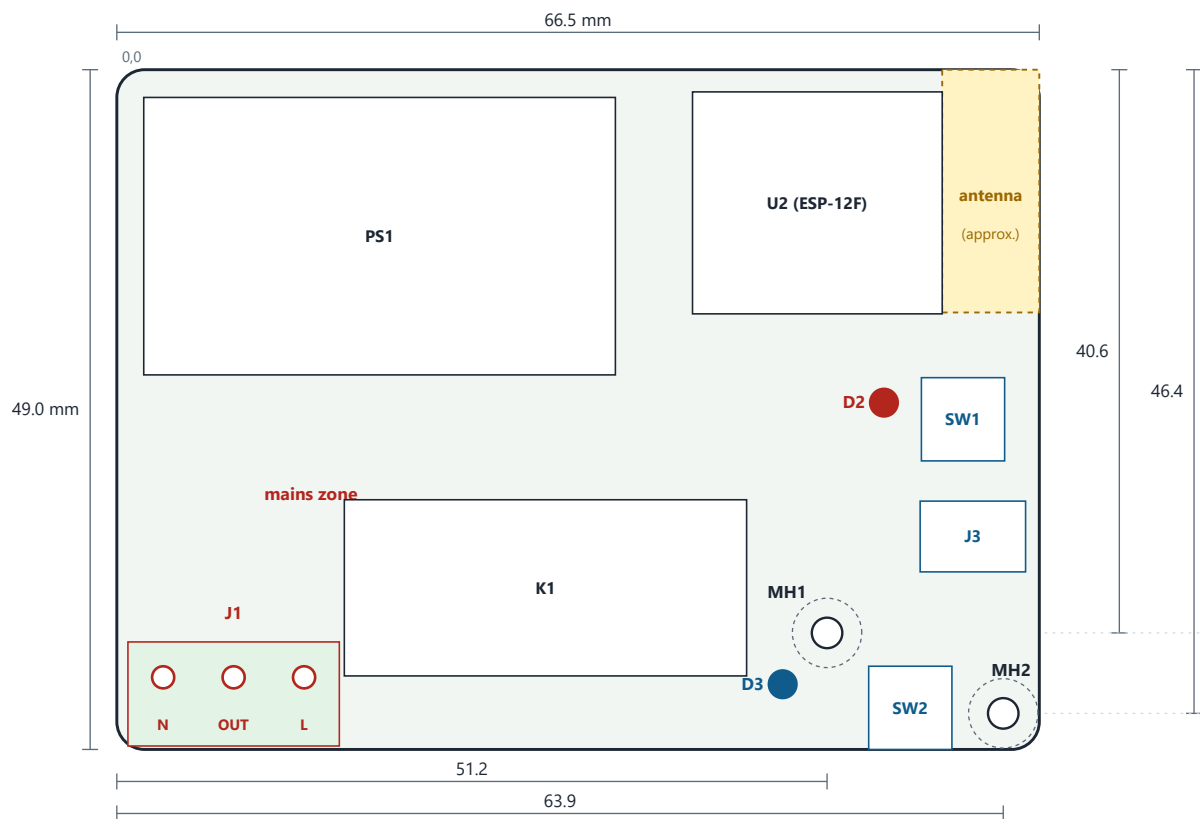
Load type	Recommended steady current	At 120 VAC	Rationale
Resistive (heater, heating element, oven)	≤ 10 A	≤ 1,200 W	[C] Trace limit; 16 A contact [D].
Incandescent / halogen lamp	≤ 5 A	≤ 600 W	[D] UL TV-5 rating at 120 VAC (5 A steady, cold-filament inrush).
LED drivers, switch-mode supplies (capacitive load)	≤ 5 A	≤ 600 W	[E] The HR model is tested at 100 A peak [D]; margin is left for the sum of inrush peaks of several drivers.
Motors, fans, transformers (inductive)	≤ 3 A	≤ 360 VA	[E] The datasheet gives no motor rating for the -HR model. Consider an RC snubber at the load.

The load is not protected by F1

F1 (T500 mA) only protects the PS1 supply. The OUT terminal must be protected upstream by a circuit breaker or fuse of **10 A maximum**. An on-board fuse would not do: its breaking capacity (tens of amperes) is far below the short-circuit current of a mains outlet.

2.5 Mechanical and environmental characteristics

Parameter	Value	Source
Board dimensions	66.5 × 49.0 mm, R2 mm corners	[M]
Thickness / material	1.6 mm, FR-4, 2 layers, 35 µm (1 oz) copper on both sides	[M]
Height above the top side	≈ 16 mm (relay 15.7 mm max. [D]; HLK-PM01 ≈ 15 mm [F])	[E]
Height below the board	≈ 2 mm (THT leads trimmed after wave soldering)	[E]
Mounting holes	2 × Ø2.2 mm (M2) at (51.2; 40.6) and (63.9; 46.4) mm; head or washer ≤ Ø5 mm	[M]
Terminal block J1	3 poles, 5.08 mm pitch, 250 V / 18 A, wire entry towards the board edge	[F]
Relay weight	≈ 10 g	[D]
Vibration (relay)	10–55 Hz, 1.5 mm double amplitude	[D]
Shock (relay)	100 m/s² malfunction; 1,000 m/s² destruction	[D]



Origin at the top-left corner (top view). R2 mm corners. MH1/MH2 holes Ø2.2 mm (M2); dashed circle = Ø5 mm keep-out for traces.

Figure 3. Mechanical drawing and interface elements (top view, dimensions in mm). The antenna area is approximate; it is drawn from the ESP-12F footprint outline.

3. Functional description

3.1 Mains input and protection

The line conductor enters through J1-3 (**L**) and the neutral through J1-1 (**N**). Two paths leave L: the load path, which goes straight to the K1 contact (net L_IN), and the supply path, which goes through fuse F1 (net L_F). Varistor RV1 sits between L_F and N, i.e. *after* the fuse.

- **Why RV1 is after F1.** Under a sustained overvoltage (e.g. 230 V connected by mistake) the varistor conducts continuously. This way the fuse opens, instead of the varistor overheating with no current limit.
- **Why F1 is time-lag (T).** The HLK-PM01 draws an inrush current when charging its input capacitor. A 500 mA fast-acting fuse could blow at every power-up.
- **Why the load does not go through F1.** The load current (up to 10 A) has nothing to do with the supply current (tens of mA). Its protection belongs to the installation (§2.4).

3.2 Isolated supply and regulation

PS1 (Hi-Link HLK-PM01) converts mains into isolated 5 V. Together with K1, it is the only part that crosses the isolation barrier. The 5 V rail feeds the relay coil and the linear regulator U1 (AMS1117-3.3), which generates the 3.3 V for the ESP-12F.

Component	Value	Function
C1	470 μ F 10 V electrolytic	5 V energy reservoir: absorbs the current step when the relay closes (80 mA) and the transmit peaks.
C2 · C3	22 μ F X5R 0805 · 100 nF	U1 input: stability and fast response.
C4 · C6	22 μ F X5R 0805 · 100 nF	U1 output. The AMS1117 needs $\geq 22 \mu$ F at its output to be stable.
C9 · C5	10 μ F X5R 0603 · 100 nF	Next to the ESP-12F VCC pin (combination recommended by Espressif). C4 is ~ 25 mm of trace away and cannot cover the transmit edges.

LDO selection rationale. With 5 V in and 3.3 V out there is 1.7 V of headroom, above the AMS1117 dropout (1.1 V typ. at 0.8 A [D]). Average dissipation is $(5 - 3.3) \text{ V} \times 80 \text{ mA} \approx 0.14 \text{ W}$; with $\theta_{JA} \approx 90 \text{ }^\circ\text{C/W}$ the junction temperature rise is about 13 $^\circ\text{C}$ (26 $^\circ\text{C}$ at transmit peak). A switching converter would save $\sim 0.1 \text{ W}$, but would add noise next to the radio and more parts.

3.3 Microcontroller and boot configuration

The ESP8266EX samples three strapping pins on the rising edge of reset to decide where to boot from. The on-board resistors select normal boot from flash; SW1 forces UART flashing mode.

Pin	Net	Components	Level at boot	Effect
GPIO15	/GPIO15	R5 10 k to GND	0	Must be 0 to boot from flash or UART.
GPIO2	/GPIO2	R4 10 k to 3V3	1	Must be 1.
GPIO0	/GPIO0	R3 10 k to 3V3; SW1 to GND; J3-5	1 (0 with SW1)	1 = boot from flash; 0 = UART flashing mode.
EN (CH_PD)	/EN	R1 10 k to 3V3; C7 100 nF to GND	$\uparrow$ with $\tau = 1 \text{ ms}$	Enables the chip once VCC has settled.
RST	/RST	R2 10 k to 3V3; C8 100 nF; SW2 to GND; J3-6	$\uparrow$ with $\tau = 1 \text{ ms}$	External reset; SW2 restarts the module.

The RC delay on EN and RST follows the Espressif power-up sequence: EN and RST must rise after VDD33 ($t_3, t_5 \geq 0.1 \text{ ms}$ [D]). Without capacitors, a slow ramp of the 3.3 V rail can leave the chip in an undefined state.

3.4 Relay driver

GPIO5 (net /RELAY) drives the base of Q1 (SS8050, NPN) through R6 (1 k Ω). Q1 switches the low side of the coil (/COIL_N); the other end is at +5 V. D1 (1N4148W) recirculates the coil current when Q1 turns off and clamps its collector to $\approx 5.7 \text{ V}$. R7 (10 k Ω) holds the base at GND while GPIO5 is high-impedance (reset, boot, flashing): the relay **cannot close** until the firmware commands it.

Quantity	Calculation	Result
Coil current	$5 \text{ V} / 62.5 \Omega$	80 mA
Base current ($V_{OH} = 3.3 \text{ V}$)	$(3.3 - 0.75) / 1 \text{ k} - 0.75 / 10 \text{ k}$	2.48 mA
Base current ($V_{OH} \text{ min.} = 2.64 \text{ V}$)	$(2.64 - 0.75) / 1 \text{ k} - 0.75 / 10 \text{ k}$	1.82 mA
Forced β (worst case)	$80 \text{ mA} / 1.82 \text{ mA}$	$\approx 44 \rightarrow$ saturation with ample margin (SS8050 $h_{FE} \square 44$)
Total GPIO5 current	$I_B + I_{D2} \approx 2.5 + 1.4$	$\approx 3.9 \text{ mA} (< 12 \text{ mA [D]})$

With a plain parallel diode the coil current decays slowly and the contact opens somewhat later than the 5 ms in the datasheet (not measured). For a 10 A AC load this is acceptable; a zener in series with D1 would speed up release at the cost of more voltage on Q1.

3.5 Indicators and push buttons

Ref.	Element	Connection	Logic	Notes
D2	Red "relay" LED	GPIO5 → D2 → R8 1 k → GND	Active high	In parallel with the driver: on = close command issued. ≈ 1.4 mA [E].
D3	Blue "Wi-Fi" LED	GPIO4 → R9 220 Ω → D3 → GND	Active high	Meaning defined by the firmware. Current ≈ 1 –2 mA, depends on the blue LED V_F [E].
SW1	TS-1187A push button	GPIO0 ↔ GND	Active low	Local switching in service; held during reset = flashing mode.
SW2	TS-1187A push button	RST ↔ GND	Active low	Module reset.

4. Connectors and pin assignment

4.1 J1 — Mains terminal block

Pin	Label	Net	Function	Recommended wire size
1	N	/N	Input neutral; shared by the load and PS1	1.5 mm ² (≈ 14 AWG) for 10 A [E]
2	OUT	/L_OUT	Switched line to the load	1.5 mm ²
3	L	/L_IN	Input line (from the circuit breaker)	1.5 mm ²

KANGNEX WJ500V-5.08-3P terminal block (250 V / 18 A) [F]. The wire entry faces the bottom edge of the board. Use copper conductors; for stranded wire, use ferrules of the right size.

4.2 J3 — Programming (2×3, 2.54 mm pitch)

IO0 pin 5	TX pin 3	GND pin 1 ■
RST pin 6	RX pin 4	5V pin 2

Pin	Signal	Connect to adapter
1 (square)	GND	GND
2	+5 V (input)	5 V / VBUS
3	ESP TXD	adapter RX
4	ESP RXD	adapter TX
5	GPIO0	GND to flash (or use SW1)
6	RST	optional (or use SW2)

Top view, with the board oriented as in Figure 3 (matches the silkscreen label). TX/RX are **3.3 V** lines: the adapter must use 3.3 V logic even though the board is powered from its 5 V.

Use J3 only with the board disconnected from mains

J3 is in the SELV zone, but an insulation failure in PS1 or K1 would bring it to mains potential, and through the adapter the computer as well. Never connect anything to J3 while mains is connected to J1.

4.3 ESP-12F pin assignment

Module pin	GPIO	Net	Direction	Use
20	GPIO5	/RELAY	Output	Relay K1 (via Q1) and LED D2. Active high.
19	GPIO4	/WIFI_LED	Output	LED D3. Active high.
18	GPIO0	/GPIO0	Input	Push button SW1 (active low, pull-up R3). Strapping pin.
22	GPIO1 / TXD	/TXD	Output	UART0 TX → J3-3. Firmware log at 115,200 bit/s (typical).
21	GPIO3 / RXD	/RXD	Input	UART0 RX ← J3-4.
17	GPIO2	/GPIO2	—	Pull-up R4. Strapping pin. Usable by the firmware with care (on many ESP-12F modules it drives the on-module blue LED, active low [F]).
16	GPIO15	/GPIO15	—	Pull-down R5. Strapping pin. Do not use.
1 · 3	RST · EN	/RST · /EN	Input	Reset (SW2, J3-6) and enable, with a 1 ms RC.
4, 5, 6, 7	GPIO16, 14, 12, 13	—	—	Not connected. Only reachable by soldering to the module castellations.
2	ADC (TOUT)	—	—	Not connected.
9–14	GPIO6–11	—	—	Internal flash bus. Do not use.

5. Hardware design

This section summarizes the engineering criteria that define the topology, layout and margins of the board. The full schematic is on the following page.

5.1 Board partitioning

- **Top strip:** PS1 on the left and the ESP-12F rotated 270° with its antenna on the right edge, over a copper-free area (footprint keep-out).
- **Bottom left, mains zone:** J1 on the bottom edge, F1 and RV1 above it and K1 to its right. No GND plane or low-voltage trace enters this zone.
- **Bottom right, low voltage:** U1 and its capacitors, C1, relay driver, push buttons, LEDs, J3 and the two mounting holes.
- **GND planes** on both sides, clipped by a polygon that geometrically stays ≥ 4.5 mm from all mains copper. The clearance therefore does not rely solely on correctly assigned net classes.
- **Mounting holes** inside the low-voltage zone, with a $\varnothing 5$ mm circle free of traces and signal vias (GND plane only), so a metal washer cannot damage a trace.

5.2 Design and isolation rules

Rule	Value	Rationale
Mains ↔ low voltage	4.0 mm (measured: 4.35 mm)	Custom rule in rele-esp12f.kicad_dru, with no exceptions across the board. Design distance between mains and SELV circuits.
Between mains nets (L_IN, L_OUT, L_F, N)	2.0 mm	Functional insulation between 120 V mains conductors.
Mains copper ↔ board edge	≥ 2.0 mm	Margin against contact with an enclosure or bracket; the enclosure must still be insulating.
K1 coil ↔ contact	8 mm (internal) / 20 mm between pads	G5RL reinforced insulation [D]. The 2 mm exception required by the previous revision's relay was removed.
SELV signal / power	0.25–0.3 mm / 0.5–0.6 mm	Standard JLCPCB manufacturing capability.

5.3 Power traces

The load current flows through L_IN (J1-3 → K1) and L_OUT (K1 → J1-2). Both are 2.5 mm traces **duplicated on both sides** and joined at the THT pads of J1 and K1. Per IPC-2221 (external layer, 35 μ m), 2×2.5 mm carry ≈ 12 A with a 20 °C rise; at 10 A the expected rise is ≈ 12 °C [C]. The G5RL-1A-E is a high-capacity model: each contact comes out on **two** pins and Omron requires both to be used [D]; on the board each pair is joined by the power trace. N, L_F and the branch to the fuse are 1.0 mm, enough for the 500 mA maximum of PS1.

5.4 Radio frequency

The ESP-12F printed antenna overhangs towards the right edge with no copper underneath on any layer. A +3V3 branch that ran next to the keep-out and split the GND plane under the module was removed. J3 is ≈ 10 mm from the antenna. In the installation, the enclosure and wiring must not cover that area (§11.2).

5.5 Copper views

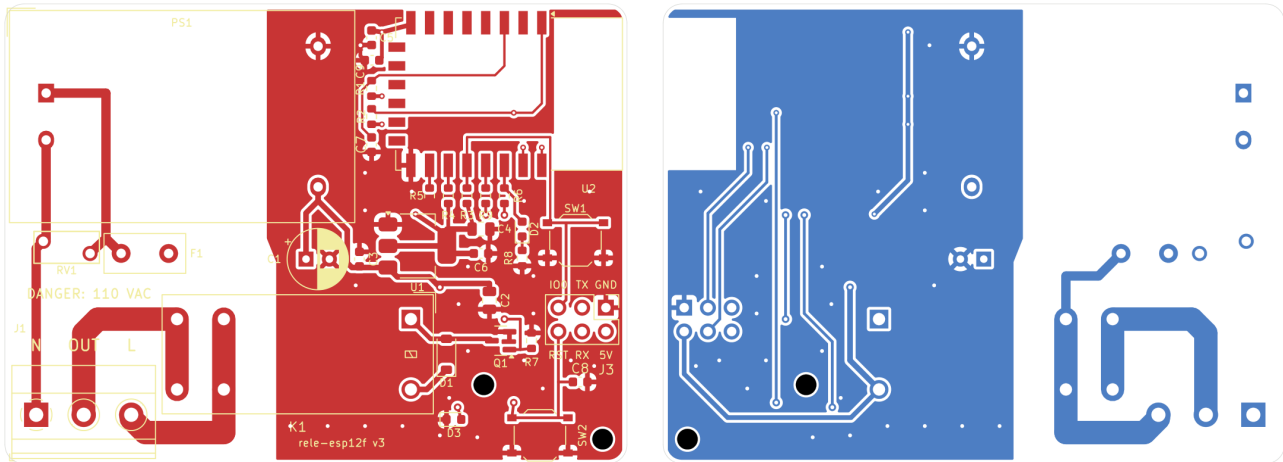


Figure 4. Left: top copper (F.Cu) with silkscreen. Right: bottom copper (B.Cu) seen from below (mirrored). The wide traces in the mains zone are L_IN and L_OUT, duplicated on both sides.

5.6 Manufacturing and assembly

- Goal: fully assembled board from JLCPCB; all 32 parts, SMD and THT, are soldered at the factory (THT by wave). No part is left for hand soldering (MH1 and MH2 are just holes).
- Files in fabrication/: gerbers and drill files (11 files), BOM_JLCPCB.csv and CPL_JLCPCB.csv, packaged in gerbers_JLCPCB.zip.
- The CPL uses the **pad center**, not the body center: this is the reference JLCPCB uses. On U2 the antenna offsets the body (X = 149.65 mm); K1 is placed at the center of its 6 pads.
- Drill sizes matched to the real parts: J1 Ø1.5 mm and RV1 Ø1.0 mm, to give insertion and wave-soldering clearance. Copper pads are unchanged, so isolation distances are too.
- Before paying, check in the JLCPCB viewer that U2 and K1 sit on their pads, that the J1 wire entry faces the edge, that J3 pin 1 lands on the square pad (GND) and that every BOM line has a part selected.

5.7 Schematic

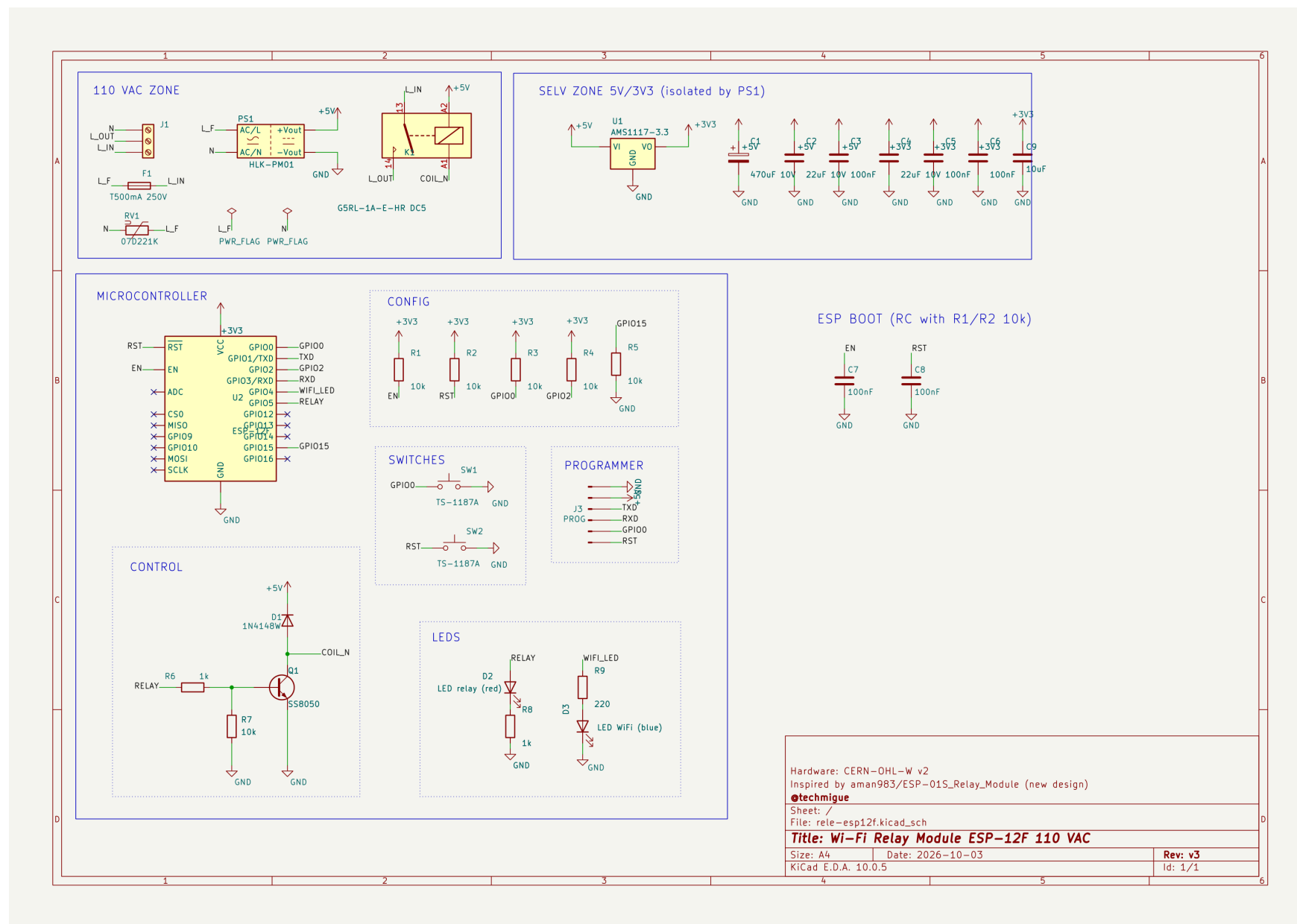


Figure 5. Full schematic (rele-esp12f.kicad_sch, exported with kicad-cli 10.0.5).

6. Bill of materials

Matches fabrication/BOM_JLPCPB.csv. All parts are mounted on the top side.

Ref.	Qty	Value / part	Package	LCSC
U2	1	ESP-12F (ESP8266EX)	SMD module	C82891
PS1	1	Hi-Link HLK-PM01, isolated AC/DC 5 V 3 W	THT	C209903
K1	1	Omron G5RL-1A-E-HR DC5, SPST-NO 16 A 250 VAC	THT	C113250
U1	1	AMS1117-3.3, 1 A LDO	SOT-223	C6186
Q1	1	SS8050, NPN	SOT-23	C2150
D1	1	1N4148W	SOD-123	C81598
F1	1	Reomax MTS0500A, T500 mA 250 V, time-lag	THT radial	C2762401
RV1	1	Varistor 07D221K	THT 7 mm disc	C49072913
J1	1	KANGNEX WJ500V-5.08-3P terminal block, 250 V 18 A	THT 5.08 mm	C72334
J3	1	Header 2×3, 2.54 mm	THT	C65114
C1	1	470 µF 10 V electrolytic	THT radial Ø6.3 mm	C112505
C2, C4	2	22 µF 10 V	0805	C45783
C9	1	10 µF 10 V X5R (Samsung CL10A106KP8NNNC)	0603	C19702
C3, C5, C6, C7, C8	5	100 nF	0603	C14663
R1–R5, R7	6	10 kΩ	0603	C25804
R6, R8	2	1 kΩ	0603	C21190
R9	1	220 Ω	0603	C22962
D2	1	Red LED (relay)	0603	C2286
D3	1	Blue LED (Wi-Fi)	0603	C51933294
SW1, SW2	2	XKB TS-1187A push button	SMD	C318884
MH1, MH2	2	M2 mounting hole (no part)	Ø2.2 mm	—

Total: 32 assembled parts (20 BOM lines) plus 2 mounting holes.

7. Design verification

Check	Tool	Result	Comment
DRC (design rules)	kicad-cli 10.0.5, zones refilled	0 errors · 0 unconnected · 29 warnings	23 warnings for footprints that differ from the library (adjusted J1/RV1 drills, custom 3D models). 6 for PS1/J1 silkscreen touching the rounded corners; the bodies stay inside.
ERC (schematic)	kicad-cli 10.0.5	0 errors · 9 warnings	3 off-grid endpoints and 6 warnings for symbols that differ from the library.
Schematic ↔ PCB parity	kicad-cli 10.0.5	No connection differences	34 warnings: LCSC/Assembly fields not copied to footprints and mounting holes without a symbol. None affects manufacturing.
Mains ↔ SELV clearance	Measured on the layout	4.35 mm min.	L_IN ↔ GND plane, on both sides.
Netlist ↔ this document	Exported XML netlist	Matches	Pin assignment, values and nets in sections 3 and 4 extracted from the netlist.

Pending validation on a physical board

- Boot and switching with mains connected; temperature of U1, K1 and power traces at 10 A.
- Ripple on the 3.3 V rail during transmission and when the relay switches.
- Wi-Fi range inside the final enclosure.
- Dielectric strength test between J1 and J3 (to confirm the isolation of the assembly).

Part II

User guide

Installation, configuration, operation and maintenance of the rele-esp12f module.

8. Safety

DANGER — Mains voltage: risk of death by electric shock

- Installation must be carried out by a qualified person, in accordance with local electrical regulations.
- Switch off the supply at the panel and verify the absence of voltage before touching the board or the wiring.
- With mains connected, the whole left half of the board (J1, F1, RV1, PS1, K1 contacts) is at mains potential. Do not touch the board while it is operating.
- The board must always be installed inside an **insulating** enclosure that prevents contact with any part of it.
- Never connect the USB-serial adapter to J3 while the board is connected to mains.

CAUTION — Intended use

- For 110–120 VAC mains only. At 230 VAC varistor RV1 conducts and is destroyed (and F1 opens). For 230 V, RV1 must be replaced with a 07D471K and the design reviewed.
- Maximum load 10 A resistive; less for other load types (§2.4). Protect the line upstream with a circuit breaker or fuse of 10 A maximum.
- The relay only interrupts the line conductor. With the relay open, the load remains connected to neutral: to work on the load, switch off at the panel.
- Do not use in safety applications, medical equipment, systems whose failure could harm people, or outdoors without a suitable enclosure.
- The assembly is not certified. Its use is the responsibility of the installer.

8.1 Board symbols and labels

Silkscreen label	Meaning
DANGER: 110 VAC	Marks the mains zone.
N · OUT · L (next to J1)	Neutral, switched output, input line.
IO0 TX GND / RST RX 5V (next to J3)	Programming header pinout.
rele-esp12f v3	Hardware revision identifier.

9. Required material

For	Material
First flashing	USB-serial adapter with 3.3 V logic and a 5 V output (CH340, CP2102, FT232...); 5 female dupont wires; PC with esptool (Python) or the Tasmota/ESPHome web flasher.
Installation	Insulating enclosure; 2 nylon M2 screws, or metal ones with a washer $\leq \varnothing 5$ mm, and standoffs; 1.5 mm ² copper wire (or the size required by the installation); ferrules for stranded wire; 3 mm flat screwdriver; voltage tester.
Line protection	Circuit breaker (or fuse) of 10 A maximum, dedicated or shared with the load.

10. First firmware flashing

The board ships **without application firmware**: the project repository does not include any. The first flashing is done through J3; later ones over Wi-Fi (OTA). This section describes the wiring and two common firmwares. The example configurations are derived from the pin assignment (§4.3) and **have not been tested on a physical board**.

10.1 Adapter connection

USB-serial adapter	→	Board J3
GND	→	pin 1 · GND (square pad)
5 V (VBUS)	→	pin 2 · 5V
TX	→	pin 4 · RX
RX	→	pin 3 · TX
(optional) GND on a separate wire	→	pin 5 · IO0, instead of pressing SW1

Why 5 V and not 3.3 V. The 3.3 V pin of a typical adapter supplies on the order of 50–100 mA, while the ESP8266 draws peaks of several hundred mA when calibrating the radio: flashing that way causes resets. With the USB 5 V (up to 500 mA) the board runs as in service, through its own regulator. Besides, feeding 3.3 V into the AMS1117 output would forward-bias its internal diode and load the whole 5 V rail.

10.2 Entering flashing mode

1. With the board **disconnected from mains**, wire the adapter as in the table above and plug it into the PC.
2. Press and hold **SW1** (IO0).
3. Press and release **SW2** (RST).
4. Release SW1. The ESP8266 now waits for data on the UART. The red LED stays off and the relay does not move.

10.3 Flashing with esptool

```
pip install esptool
esptool.py --chip esp8266 --port COM5 --baud 115200 flash_id      # check connection and flash size
esptool.py --chip esp8266 --port COM5 --baud 115200 erase_flash # repeat steps 2-4 first
esptool.py --chip esp8266 --port COM5 --baud 460800 write_flash -fm dout 0x0 firmware.bin
```

Replace COM5 with the adapter port (on Linux, /dev/ttyUSB0). Flashing mode must be entered again (steps 2–4) before each command, because this header has no DTR/RTS auto-reset circuit. When done, press SW2 to start the firmware.

10.4 Option A: Tasmota

Flash tasmota.bin (generic ESP8266 build) and apply this template under *Configuration* → *Configure Other* → *Template*, ticking *Activate*:

```
{"NAME": "rele-esp12f", "GPIO": [32, 0, 0, 0, 544, 224, 0, 0, 0, 0, 0, 0, 0], "FLAG": 0, "BASE": 18}
```

Position	GPIO	Code	Tasmota function
1	GPIO0	32	Button1 (SW1)
5	GPIO4	544	LedLink (D3, shows Wi-Fi/MQTT connection)
6	GPIO5	224	Relay1 (K1 and D2)
others	—	0	Unused (GPIO1/3 remain the UART for logging)

Recommended console settings: PowerOnState 0 (relay open after a power outage) or PowerOnState 3 (restore the last state), and SetOption1 1 so that a long press on SW1 does not trigger an accidental factory reset. Check the codes against the documentation of the Tasmota version used.

10.5 Option B: ESPHome

```

esphome:
  name: rele-esp12f
esp8266:
  board: esp12e          # ESP-12F: same pinout and 4 MB flash
  restore_from_flash: true
wifi:
  ssid: !secret wifi_ssid
  password: !secret wifi_password
  ap: {}                  # fallback access point if it cannot connect
captive_portal:
logger:
api:
ota:
  - platform: esphome
status_led:
  pin: GPIO4             # D3, active high
switch:
  - platform: gpio
    id: relay
    name: "Relay"
    pin: GPIO5            # Q1 -> K1, and LED D2
    restore_mode: RESTORE_DEFAULT_OFF
binary_sensor:
  - platform: gpio
    name: "Button SW1"
    pin:
      number: GPIO0
      inverted: true       # active low, external pull-up R3
    filters:
      - delayed_on: 30ms   # debounce
    on_press:
      - switch.toggle: relay

```

Custom firmware (Arduino / PlatformIO)

- Board: "Generic ESP8266 Module" or "NodeMCU 1.0 (ESP-12E)"; 4 MB flash, DOUT or DIO mode.
- Configure GPIO5 as an output and write LOW first thing in `setup()`.
- Read GPIO0 as a debounced input. Do not use GPIO6–11, and do not turn GPIO15/GPIO2 into outputs that could be left at the wrong level on reset.
- Include OTA updates from the first version: once the board is installed, J3 can no longer be used.

11. Installation

11.1 Wiring diagram

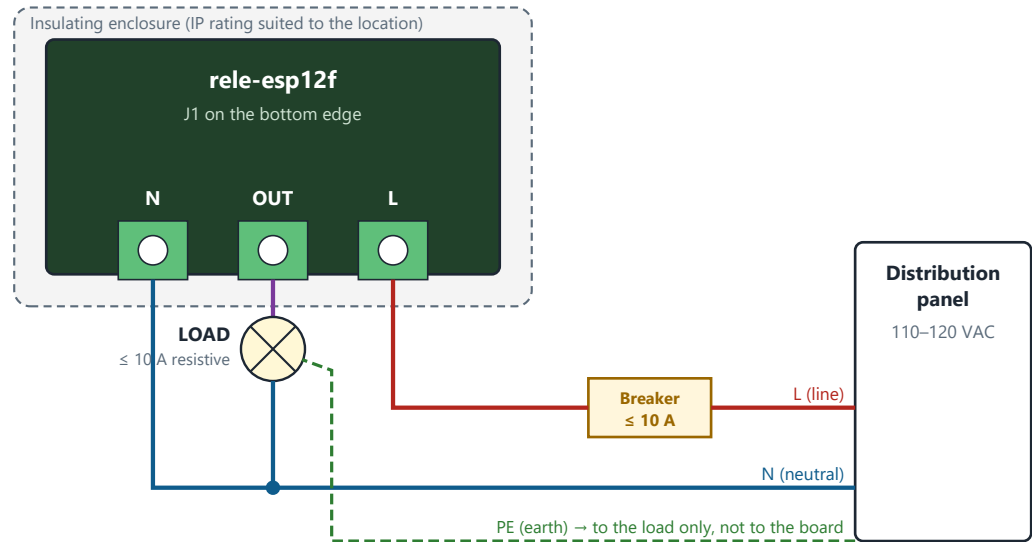


Figure 6. Typical wiring. The relay interrupts the line conductor to the load; the neutral is shared. The protective earth (PE) goes directly to the load.

11.2 Mechanical mounting

- Choose a **plastic** (non-metallic) enclosure with an IP rating suited to the location and enough inner space for the board ($66.5 \times 49 \times \approx 20$ mm) plus the wiring and its bend radius.
- Fix the board through holes MH1 and MH2 (M2) with standoffs of ≥ 3 mm so the THT leads do not touch the bottom. Do not pass screws through the board anywhere else.
- Use screws with a head or washer of $\varnothing \leq 5$ mm: this is the trace-free diameter around each hole.
- Keep the antenna area (top-right corner, Figure 3) free of metal and wiring. If the enclosure goes inside a metal box, the Wi-Fi range will drop significantly.
- Keep the mains wiring physically away from J3 and the ESP-12F; if needed, secure it to the enclosure with cable ties.
- Provide passive ventilation if the load approaches 10 A: the power traces, the relay and the terminal block heat up.

11.3 Electrical connection

1. Switch off the circuit at the panel and **verify the absence of voltage** with a suitable tester.
2. Strip 6–7 mm of each conductor (or crimp ferrules on stranded wire) [E: verify with the terminal block datasheet].
3. Connect the **line** coming from the circuit breaker to **L** (J1-3).
4. Connect the **neutral** to **N** (J1-1). If the load shares the neutral, join it off-board with a splice connector; J1 accepts a single conductor per pole [E].
5. Connect the line wire going to the load to **OUT** (J1-2). The other load pole goes to neutral; earth goes directly to the load.
6. Tighten the J1 screws firmly (≈ 0.4 – 0.5 N·m, typical for 5.08 mm terminal blocks; check the KANGNEX datasheet [F]). Gently pull each wire to verify it.
7. Check there are no loose strands or exposed copper outside the terminal block, and close the enclosure before applying power.

11.4 Pre-commissioning checks

#	Check	Correct if...
1	Firmware flashed and tested while powered from J3 (no mains)	The blue LED shows the connection and the relay switches with SW1 / over Wi-Fi.
2	J3 disconnected and free of wires	Nothing remains connected to the programming header.
3	J1 polarity	L on the pole labeled "L", N on "N", load on "OUT".
4	Upstream protection	Circuit breaker or fuse $\leq 10\text{ A}$ on the line.
5	Enclosure closed	No part of the board is accessible.
6	Power-up	After 1–3 s the firmware boots; the relay stays open unless the firmware restores a "closed" state.

12. Configuration

The steps depend on the firmware. This is the general sequence with Tasmota and ESPHome:

12.1 Connecting to the Wi-Fi network

Step	Tasmota	ESPHome
First boot	Creates the access point tasmota-XXXXXX.	If the credentials in secrets.yaml fail, it creates the fallback AP.
Configure Wi-Fi	Join the AP, open 192.168.4.1 and enter SSID and password (2.4 GHz only).	Credentials are compiled in; with the fallback AP, captive portal at 192.168.4.1.
Find the IP address	From the router, or from the serial console during flashing.	Home Assistant discovers it automatically (ESPHome integration).
Integration	MQTT (Configuration → MQTT) or the Home Assistant Tasmota integration.	Native Home Assistant API.

12.2 Recommended settings

- **State after a power outage:** explicitly choose "off" or "last state" depending on the load. For heaters or loads that must not start on their own, use "off".
- **Web UI and OTA password:** always set one. Otherwise anyone on the local network could switch the load or upload other firmware.
- **Wi-Fi network:** the ESP8266 only works on 2.4 GHz and with WPA2 (not WPA3-only). If the router merges bands under the same name, the 2.4 GHz band may need to be split out.
- **Limit switching:** avoid automations that switch the relay many times per minute; the electrical life of the contact is finite (§14.2).
- **Time (NTP) and time zone,** if timers are used.

12.3 OTA updates

Tasmota: *Firmware Upgrade* from the web UI, with a .bin or .bin.gz file. ESPHome: `esphome run rele-esp12f.yaml`, which compiles and uploads over the network. With 4 MB of flash there is room for two-stage updates. If an update fails and the module does not boot, the only option left is flashing through J3, removing the board from the installation (§10).

13. Operation

13.1 Indicators

Indicator	State	Meaning
Red LED (D2)	On	The firmware commands the relay to close (GPIO5 high). It is wired in parallel with the driver, so it reflects the command, not the mechanical position of the contact.
Red LED (D2)	Off	Relay open; also during boot and flashing.
Blue LED (D3)	Firmware-defined	Tasmota (LedLink): blinks while looking for Wi-Fi/MQTT, off once connected (configurable with LedState). ESPHome (status_led): blinks on error or warning, off when all is well.
Audible click	When switching	K1 contact closing or opening. If the red LED changes without a click, see §15.

13.2 Push buttons

Button	Action	Result
SW1	Short press in service	Toggles the relay (if the firmware implements it; it does in the configurations of §10).
SW1	Held at boot or while pressing SW2	Flashing mode: the firmware does not start. Release and press SW2 to return to normal mode.
SW1	40 s press (Tasmota)	Restores the Tasmota factory settings (unless SetOption1 1).
SW2	Press	Restarts the module. The relay opens during the restart and returns to the state defined by the firmware.

The push buttons are on the board

SW1 and SW2 are only reachable with the enclosure open, that is, next to parts at mains potential. Use them only with the board disconnected from mains (powered from J3), or add an external button on the enclosure wired with adequate insulation.

13.3 Behavior on events

Event	Board behavior
Power-up / mains restored	Relay open for $\approx 0.1\text{--}1$ s until the firmware starts (R7 keeps Q1 off); then the configured state.
Wi-Fi loss	The relay keeps its state. Local control (SW1) and internal timers keep working if the firmware allows it.
Power outage	The relay opens (no coil supply). The load is de-energized.
Transient overvoltage	RV1 clips the peaks at the PS1 input. It does not protect the load: use a surge protector at the panel for that.
Sustained overvoltage (e.g. 230 V)	RV1 conducts, F1 opens and the board stops working. Requires repair (§14.4).
Short circuit in the load	Must be cleared by the upstream circuit breaker. The K1 contact may be damaged or welded: inspect it (§15).

14. Maintenance

Before any intervention

Switch off the supply at the panel and verify the absence of voltage. No maintenance task is performed with the board energized.

14.1 Inspection schedule

When	Task	What to look for
One month after installation	Re-tighten the J1 screws	Copper settles after the first thermal cycles and the connection loosens.
Every 12 months	Visual inspection with the enclosure open and de-energized	Discoloration or charring at J1, around K1 or on the power traces; RV1 bulged, cracked or blackened; C1 bulged; burning smell.
Every 12 months	Functional test	Switching from the app and with SW1; audible click; the load turns on and off.
Every 12 months	Firmware	Apply firmware security updates over OTA.
After a storm or surge	RV1 inspection and functional test	A varistor that has absorbed strong surges degrades; if damaged, repair it (§14.4).
If the load runs close to 10 A	Check every 6 months	Terminal block and relay temperature; if possible, thermography with the load connected (from outside the enclosure).

14.2 Relay service life

Omron guarantees a minimum of **50,000 operations** at 16 A resistive and 250 VAC [D]. At lower currents the life is longer (endurance curve in the datasheet). As a conservative reference:

Operations per day	Estimated minimum life at full load [C]
10	≈ 13.7 years
20	≈ 6.8 years
50	≈ 2.7 years
200 (aggressive automation)	≈ 8 months

End-of-life symptoms: the load does not turn off (welded contact), does not turn on or flickers (burnt contact), or there is abnormal heating around K1. In those cases, replace the relay or the board.

14.3 Cleaning

De-energized, with dry air or an antistatic brush. Do not use water or solvents, and do not apply conformal coating on the relay or the terminal block. If moisture has entered or there is condensation in the enclosure, keep the board out of service until it is completely dry and check the enclosure sealing.

14.4 Repairs

- **F1 is not a user-replaceable fuse:** it is soldered. If it opens, the cause is usually an overvoltage (RV1) or a PS1 fault. Diagnose it before replacing the fuse, always with the same type: T500 mA 250 V, time-lag (Reomax MTS0500A or an equivalent with the same footprint).
- RV1, K1 and PS1 are THT and can be replaced with a soldering iron and a desoldering tool. Use exactly the BOM part numbers (§6): a relay without the "-E" variant or a different supply changes the current rating or the isolation.
- After any repair in the mains zone, make sure no solder residue or wire strands reduce the isolation distances, and repeat the checks of §11.4.

15. Troubleshooting

For any measurement with the board open, power it from J3 with the USB adapter (5 V) and **with no mains on J1**. This allows checking the whole SELV section safely. Faults in the mains zone must be diagnosed de-energized (continuity, resistance) or by a qualified technician.

Symptom	Probable cause	Check / solution
With mains: no LED, not visible on the Wi-Fi network	F1 open, PS1 faulty, terminal block badly wired or no firmware.	Without mains, power from J3: if it works, the fault is in the mains zone (F1, RV1, PS1). Check F1 continuity de-energized.
Firmware does not start; the serial console shows "boot mode:(1,x)"	GPIO0 low at boot.	SW1 stuck or IO0 wire to GND on J3. Release it and press SW2.
esptool does not connect ("Failed to connect")	Not in flashing mode; TX/RX not crossed; port or drivers.	Repeat §10.2; cross TX↔RX; try 115,200 bit/s; check the COM port.
Resets when Wi-Fi connects or the relay switches	Insufficient supply.	Via J3: use the adapter 5 V, not 3.3 V, and a short USB cable. With mains: check C1/C9 and the PS1 output.
Red LED changes but there is no click and no switching	Q1, D1 or K1 coil faulty; low 5 V.	Measure 5 V between J3-2 and J3-1 (powered from J3). With GPIO5 high, the Q1 collector must be < 0.3 V.
Click heard but the load gets no voltage	OUT/N wiring, breaker open, burnt contact.	De-energized, check L↔OUT continuity with the relay closed (powered from J3).
The load never turns off	K1 contact welded (inrush or short circuit).	De-energized: L↔OUT continuity with the relay at rest. If there is continuity, replace K1 and check the load.
Switches but the red LED does not light	D2 or R8 faulty or badly soldered.	No functional impact. Check D2/R8.
Weak Wi-Fi or disconnections	Metal enclosure, covered antenna, distance, router on 5 GHz.	Relocate the enclosure; clear the antenna area; 2.4 GHz network; repeater.
Hot terminal block or enclosure	Loose screw, undersized wire, load > 10 A.	Re-tighten, 1.5 mm ² wire, reduce the load. If there is discoloration, replace the board.
Does not work after a storm	RV1 and F1 damaged by overvoltage.	Visual inspection of RV1; repair per §14.4.

Appendix A. Design calculations

A.1 Power budget

Consumer	Rail	Typical	Worst case	Source
ESP8266EX (average / 802.11b TX)	3.3 V → 5 V via U1	80 mA	170 mA	[D]
LED D3	3.3 V	≈ 2 mA	≈ 3 mA	[E]
LED D2 + Q1 base	3.3 V	3.9 mA	3.9 mA	[C]
Dividers and pull-ups (R1–R5, R7)	3.3 V	< 0.5 mA	< 1 mA	[C]
K1 coil	5 V	80 mA	88 mA (–10 % R)	[D]
Total on 5 V		≈ 166 mA	≈ 266 mA	[C]
Margin on PS1 (600 mA)		72 %	56 %	[C]

The radio calibration peaks at boot (hundreds of mA for milliseconds) are covered by C1, C4 and C9. Power drawn from mains, assuming a PS1 efficiency of 70 % [E], is ≈ 0.6 W with the relay open (≈ 86 mA × 5 V / 0.7) and ≈ 1.2 W with the relay closed (≈ 166 mA × 5 V / 0.7).

A.2 Regulator U1

$P_{U1} \text{ (average)} = (5.0 - 3.3) \text{ V} \times 0.080 \text{ A} = 0.136 \text{ W} \rightarrow dT_j = 0.136 \times 90 \text{ C/W} \sim 12 \text{ C}$
 $P_{U1} \text{ (peak)} = (5.0 - 3.3) \text{ V} \times 0.170 \text{ A} = 0.289 \text{ W} \rightarrow dT_j = 0.289 \times 90 \text{ C/W} \sim 26 \text{ C}$
 Dropout margin: $5.0 - 3.3 = 1.7 \text{ V} > 1.1 \text{ V typ. (0.8 A) / 1.3 V max.}$

A.3 Relay driver

$I_{\text{coil}} = 5 \text{ V} / 62.5 \text{ ohm} = 80 \text{ mA}$
 $I_B = (V_{OH} - V_{BE}) / R6 - V_{BE} / R7 = (2.64 - 0.75) / 1000 - 0.75 / 10000 = 1.82 \text{ mA}$ (minimum $V_{OH} = 0.8 \times 3.3 \text{ V}$)
 $\beta_{\text{forced}} = 80 / 1.82 \sim 44$ (saturation guaranteed)
 $V_{\text{coil}} = 5.0 - V_{CE(sat)} (\sim 0.2 \text{ V}) \sim 4.8 \text{ V} > 3.5 \text{ V (70 \% , must operate)}$
 $V_{\text{collector at turn-off}} = 5.0 + V_F(D1) (\sim 0.7 \text{ V}) \sim 5.7 \text{ V}$

A.4 EN and RST delay

$\tau = R1 \times C7 = 10 \text{ kohm} \times 100 \text{ nF} = 1 \text{ ms}$
 $t(V = 0.75 \times 3.3 \text{ V}) = -\tau \times \ln(1 - 0.75) = 1.39 \text{ ms}$

A.5 Power trace capacity (IPC-2221, external layer)

$I = k \times dT^{0.44} \times A^{0.725}$ $k = 0.048$ (external), A in mil², dT in C
 $2.5 \text{ mm} \times 35 \text{ um trace} \rightarrow A = 98.4 \text{ mil} \times 1.38 \text{ mil} = 136 \text{ mil}^2$
 $dT = 20 \text{ C: } I(1 \text{ trace}) \sim 6.3 \text{ A} \quad I(2 \text{ traces, F.Cu} + \text{B.Cu}) \sim 12 \text{ A}$
 $I = 10 \text{ A over 2 traces (5 A each): } dT \sim 12 \text{ C}$

A.6 Relay life

Life (years) = $50,000 \text{ operations} / (\text{operations_per_day} \times 365)$
 $20 \text{ operations/day} \rightarrow 50,000 / 7,300 = 6.8 \text{ years}$ (guaranteed minimum at 16 A resistive)

Appendix B. References

Document	Location
Omron G5RL PCB Power Relay, Cat. No. K132-E1-10	docs/datasheets/omron_g5rl.pdf
AMS1117 1A Low Dropout Voltage Regulator	docs/datasheets/ams1117.pdf
ESP8266EX Datasheet (2025.11, NRND)	docs/datasheets/0a-esp8266ex_datasheet_en.pdf
ESP8266 Hardware Design Guidelines	docs/datasheets/esp8266_hardware_design_guidelines_en.pdf
ESP-WROOM-02 Datasheet (module design reference)	docs/datasheets/esp-wroom-02_datasheet_en.pdf
Hi-Link HLK-PM01 (not included)	Manufacturer datasheet; LCSC C209903
Schematic and layout	rele-esp12f.kicad_sch / rele-esp12f.kicad_pcb (KiCad 10)
Manufacturing files	fabrication/ (BOM, CPL, gerbers, ZIP for JLCPCB)
IPC-2221B, Generic Standard on Printed Board Design	Trace current capacity

Appendix C. Known open points

Items identified while preparing this document. None prevents manufacturing the board.

#	Where	Open point	Approach taken here
1	docs/datasheets	The datasheets for HLK-PM01, KANGNEX WJ500V, 07D221K, SS8050 and Reomax MTS are missing.	Their values are tagged [F].
2	Espressif	The ESP8266EX is listed as NRND.	Warning in §1.4.
3	Firmware	The repository does not contain firmware.	Third-party firmwares are described (§10) with untested configurations.

Appendix D. Revision history

Revision	Date	Changes
Hardware v1	2026-09-25	66 × 66 mm board with SRD-05VDC-SL-C relay.
Hardware v2	2026-09-30	66.5 × 44 mm board, 110 VAC; new layout, 2×3 J3, GND planes with an isolation polygon, single-sided assembly.
Hardware v3	2026-10-03	Omron G5RL-1A-E-HR relay (16 A, reinforced insulation); 66.5 × 49 mm board; C7/C8 delay on EN/RST; C9 10 µF next to the ESP-12F; J3 powered at 5 V; L_IN/L_OUT duplicated on both sides (10 A); mains copper ≥ 2 mm from the edge; adjusted J1/RV1 drills; full assembly at JLCPCB.
Document rev. A	2026-10-03	First edition of the datasheet and user guide (Spanish).
Document rev. B	2026-10-04	English edition; silkscreen and schematic labels updated to English.